AMAARA SUBRAMANIAM

Aditya Birla World Academy (ABWA) | Academic & Extracurricular Resume

Emerging researcher in human genetics, gene therapy, and precision medicine, with independent and co-authored publications, research placements at IIT Bombay and other research laboratories, and creator of an integrated leukodystrophy knowledge atlas connecting genetic evidence, disease mechanisms, and therapeutic research.

Research Experience

Founder & Lead Researcher, GeneMap | *Independent capstone* | *Jan 2026–present*

Mentors: Louis (Oxford); Dr. Mayuri Bandekar (IIT Bombay) · thegenemap.com

- Conceived and developed an interactive research atlas spanning 35 genetic leukodystrophies, integrating approximately 3,200 pages of cited biomedical evidence into a unified, searchable resource.
- Synthesized evidence across disease genetics, pathogenic variants, genotype–phenotype relationships, molecular and cellular mechanisms, clinical manifestations, biomarkers, diagnostic approaches, and emerging therapeutic strategies.
- Designed a cross-disease framework for comparing disorders from molecular pathogenesis to therapeutic development, including gene-replacement therapies, clinical trials, and investigational treatment pipelines.
- Linked primary literature, biomedical databases, standardized identifiers, clinical-trial records, and source-level evidence classifications to support traceability and distinguish established findings from emerging, investigational, or unresolved areas.
- Developed disease-specific research dossiers and an integrated evidence architecture to help researchers identify shared mechanisms, compare therapeutic approaches, formulate hypotheses, and uncover research gaps.
- Designed the platform's information architecture and user experience around specialist-validated, evidence-linked content.

Independent Researcher, PITRM1 Pan-Cancer Bioinformatics Analysis | *Manuscript accepted for publication*

- Conceptualized and conducted an independent investigation into the cancer-associated role of PITRM1, a mitochondrial protease implicated in protein quality control, using publicly available cancer-genomics datasets.
- Analyzed 156,010 samples from 148,790 patients across 12 pan-cancer studies to characterize PITRM1 genomic alterations, including mutations and copy-number changes, and tumor-versus-normal expression patterns.
- Conducted mutation-distribution and Kaplan–Meier survival analyses, identifying cancer-specific associations between PITRM1 alterations and poorer overall survival in brain and thyroid cancer cohorts, but no significant association in the overall pan-cancer or lung cancer analyses.
- Integrated genomic and clinical findings to investigate the potential relationship between mitochondrial proteostasis and tumor biology, highlighting context-dependent associations and avenues for further functional investigation.
- Authored "Exploring the Cancer-Associated Role of PITRM1: A Pan-Cancer Bioinformatics Analysis," accepted for publication in the Oxford Journal of Student Scholarship (OJSS).

Research Intern, Single-Cell Genomics & Computational Biology | *Saket Chaudhary, IIT* | *June 15–July 25*

- Served as the laboratory's sole high-school intern.
- Worked independently through the Seurat workflow in R, including quality control, normalization, highly variable gene selection, PCA, clustering, UMAP, differential expression, and cell-type annotation.
- Reviewed approximately 200 scientific papers and built a structured Excel database to support curation of an MASLD (fatty liver) atlas. The assigned task was to define a "healthy" liver and address inter-study heterogeneity in definitions across global laboratories.
- Attended research seminars on current genomics work.

Research Intern, Gene Therapy | *BARC Hospital* | *Dr. Rahul Checker* | *6 weeks*

- Studied plasmid design, bacterial transformation, PCR, gel electrophoresis, transfection, and protein expression and validation.
- Examined AAV vectors, capsid engineering, and oligodendrocyte-targeted delivery in the context of Canavan disease (ASPA/NAA biology) and preclinical and clinical gene-therapy development.
- Wrote a research report on these topics; received a letter of recommendation from Dr. Checker.

Publications

- Co-author, "A Student Perspective: N-Acetylaspartate (NAA) Detection by Urine GCMS Analysis in a Rare Genetic Brain Disorder – Canavan Disease." ER Publications; published. Mentor: Dr. Usha Dave. Oct 2025–Jan 2026. Awarded CREST Gold Award.
- Author, "Exploring the Cancer-Associated Role of PITRM1: A Pan-Cancer Bioinformatics Analysis." Accepted for publication in the Oxford Journal of Student Scholarship (OJSS); IIT Bombay/BARC, mentor Dr. Mayuri Bandekar.
- Contributor, MASLD atlas project, IIT; mentor Saket Chaudhary. Publication forthcoming.

Applied Projects

Founder & AI Developer, MannPakad | *AI therapy app for teens with OCD*

- Built an app that delivers clinically grounded OCD therapy to teens who cannot access a therapist, adapting to each user.
- Clinical validation in progress with psychologist Dr. Nikita and the Muskaan Foundation.
- Qualified through the first round of INSEF and advanced to INSEF Regionals in Grade 11.

Clinical & Laboratory Training

Genomics Intern | *Dr. Usha Dave, Labassure Advanced Genomics Institute* | *Grade 11*

- Trained in genetic disorder classification, chromatography, GC-MS metabolic testing, pedigree analysis, cancer biology, immunology, and sample handling.
- Completed laboratory biosafety training and one-to-one forensic testing training; visited the ACTREC cancer research laboratory (Kharghar) and biotechnology start-ups in Pune.
- Wrote a final internship report; this placement led to the Canavan diagnostics research project listed above.

Genetics Observer | *Dr. Hema Purandare, MedGenome / Centre for Genetic Health Care*

- Shadowed a pioneer of prenatal genetics across 6+ live counselling sessions on how DNA testing informs decisions for families.

Education

Aditya Birla World Academy (ABWA), Mumbai | *Grades 9–12* | *Expected graduation: 2027*

- Grade 9: English (FLE), Computer Science, Biology, Spanish.
- Grade 10 (International): Mathematics, Spanish, Computer Science, Economics, Biology, Physics, Chemistry, Global Perspectives.
- Grade 11: English General Paper (EGP), Biology, Physics, Mathematics.
- Academic Excellence awards in the subjects listed above, 2024–2026; General Proficiency, Grades 9–11.

Honors & Awards

- CREST Gold Award, 2026.
- INSEF Regionals, Grade 11.
- Immerse Education Essay Competition, Honours List, 2026.
- DELE Spanish A2, 93%, Grade 11.
- Dr. Homi Bhabha Balvaidyanik Competition, participant, 2023–24.

World Scholar's Cup — Bangkok Global Round (Aug 21, 2024)

- Gold: Overall Challenge (6th of 500 participants); Challenge: Special Area (11th); Challenge: Art & Music (15th); Collaborative Writing (50th); Scholar's Bowl, team (41st).
- Silver: Senior Team (10th); South Asia Team (5th); Overall Scholar (99th). Trophy: Team Writing (6th).

World Scholar's Cup — Mumbai Regional Round

- Champion Challenge Scholar (Jac Khor Trophy); Overall Scholar trophy; top-scoring scholar from school (gold).
- Team awards: 3rd Overall Team, Champion Challenge Team, Writing Team, and Bowl Team trophies; Debate Team silver (7th).
- Gold medals: Writer, Challenge Art & Music, Challenge History, Challenge Special Area. Silver medals: Debater (19th), Challenge Social Studies (5th).

School Honors, ABWA

- Academic Excellence, 2024–2026 (subjects listed under Education); General Proficiency, Grades 9–11.
- Attribute Awards, 2023 (Grade 9) and 2024, for empathy, diligence, resilience, honesty, respect, and responsibility.
- Muskaan Foundation Award for Social Service, Grade 11.

Leadership, Service & Activities

Founder & Program Lead, Animal-Assisted Therapy | *Yes I Can Foundation × Animal Angels Foundation India*

- Brokered a first-of-its-kind NGO partnership bringing animals to interact with and provide comfort to isolated elderly residents; ran the program to support residents' mental wellbeing.

Rescue Volunteer & Content Creator | *YODA (Youth Organisation for Defence of Animals) and WSD Initiative*

- Rescued and rehabilitated injured strays; ran social campaigns that drove adoptions and donations.

Writing & Other Activities

Core Writer, Newsletter Team | *ABWA School Newsletter* | *Grade 10–present*

- Published 20+ articles; started the newsletter's STEM column on current science, student research, and ethics in STEM.
- Competed in the Derek O'Brien parliamentary writing contest at national level.

Team Captain | *World Scholar's Cup, Bangkok Global Round*

Vocalist | *Trinity College London*

- Earned Distinction in Grades 1 and 2; performed solo and with bands for 3+ years, including a 150-person concert.

Trainee | *Krav Maga India* | *Grade 9–present*

Team Member | *Binary Bolts, FIRST Robotics Competition* | *2022–23*

- Helped design and build a competition robot under deadline.

Coursework & Skills

- Independent coursework: Stanford Online, RNA Biology; University of Chicago, Neurobiology (both via Coursera, 90%+, no credit).
- Computational: R and the Seurat workflow for single-cell RNA-seq; systematic literature review and database curation.
- Laboratory exposure: PCR, gel electrophoresis, plasmid construction, GC-MS, chromatography, sample preparation.
- Domain knowledge: Leukodystrophies, Canavan disease, AAV gene therapy, prenatal and cancer genetics, single-cell RNA sequencing.
- Languages: English; Spanish (DELE A2, 93%).